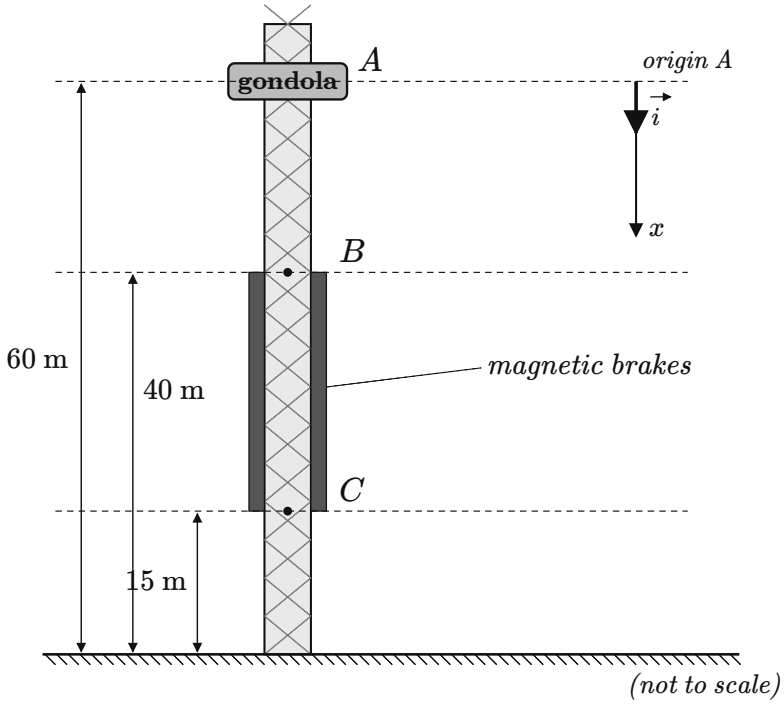
Grade 12 — Life Sciences Lebanese curriculum	**Test Preparation** Chapter 1 — Energy	Duration: 1 h 30 min Total: 20 points

This test is formed of two obligatory exercises. The use of a non-programmable calculator is allowed.

Take, in all exercises: $g = 10\ \text{m/s}^2$.

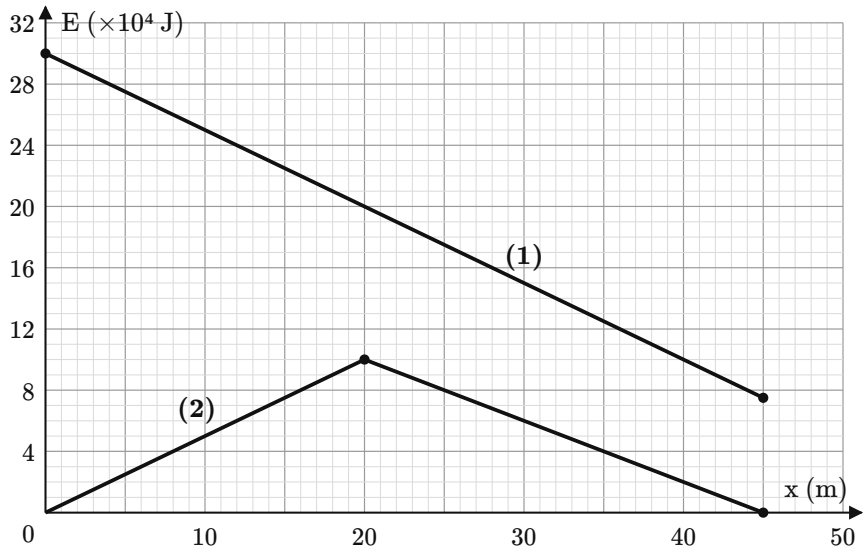
Exercise 1 — The drop tower (10 points)

In an amusement park, a gondola carrying its passengers is lifted to the top A of a vertical tower, then released from rest. It falls freely from A to B. Between B and C, magnetic brakes fixed on the tower exert on the gondola a force $\vec{F}$ of constant magnitude, opposite to its velocity, until it stops at C (Doc. 1). A data logger on board gives the energies of the system [gondola, Earth] during the descent (Doc. 2).



Doc. 1

Mass of the gondola with its passengers: $M = 500$ kg; it is considered as a particle.
Heights above the ground: $z_A = 60$ m, $z_B = 40$ m, $z_C = 15$ m.
The position of the gondola is located by its abscissa x on the vertical axis $(A, \vec{i})$ directed downward; v is the algebraic value of its velocity.
The ground is the reference level for gravitational potential energy. Air resistance is neglected in Parts 1 and 2.



Doc. 2

Part 1 — The free fall (from A to B)

1.1 Express the gravitational potential energy of the system in terms of M, g and x. Identify, with justification, the curve of Doc. 2 that represents it.

1.2 Using Doc. 2, show that the mechanical energy of the system is conserved between A and B. Justify this conservation.

1.3 By deriving the mechanical energy with respect to time, determine the acceleration a_1 of the gondola. Specify the nature of its motion.

1.4 Write the time equations $v(t)$ and $x(t)$ of the gondola between A and B. Deduce the instant t_B at which it passes through B and its speed v_B there.

Part 2 — The magnetic braking (from B to C)

2.1 Using Doc. 2, determine the variation of the mechanical energy of the system between B and C. Deduce the variation of the internal energy of the system [gondola, brakes, Earth].

2.2 Express the mechanical energy of the system between B and C in terms of x, the mechanical energy at B and F. Deduce the value of F.

2.3 By deriving the mechanical energy with respect to time, determine the acceleration a_2 of the gondola. Specify the nature of its motion.

2.4 Determine the duration of the braking, then the total duration of the descent from A to C.

2.5 The designers require that the magnitude of the deceleration of the passengers never exceeds $2g$. Is this requirement satisfied?

Part 3 — The real ride

In reality, the air exerts on the gondola a resistive force of constant magnitude f during the whole descent. The data logger now gives the kinetic energy $KE_B = 9.0 \times 10^4$ J at B.

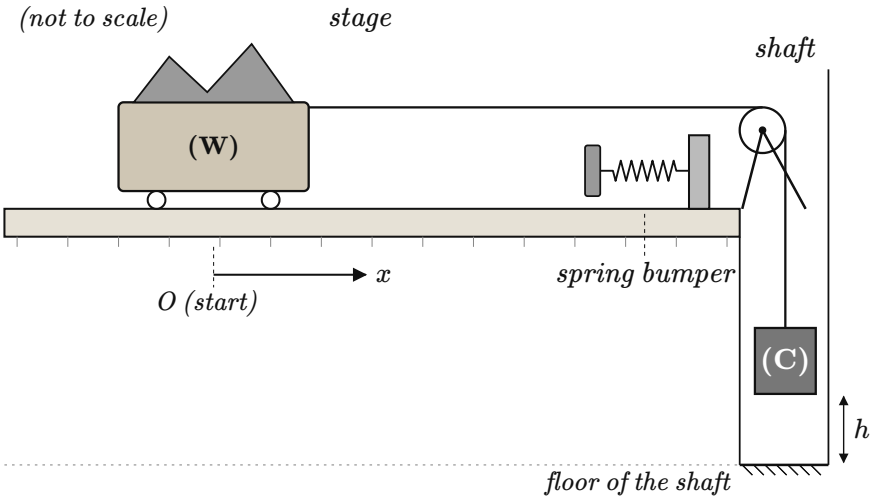
3.1 Determine f.

3.2 Determine the new acceleration of the gondola between A and B and the instant at which it reaches B ($\sqrt{360} \approx 19$).

3.3 The brakes are unchanged. Show that the gondola now stops 5.0 m above C.

Exercise 2 — Behind the curtain: a counterweight on stage (10 points)

In a theatre, a heavy scenery wagon (W) is moved across the stage by a counterweight (C) that falls in a vertical shaft. The rope passes over a pulley at the edge of the stage (Doc. 3). At $t_0 = 0$, the system is released from rest. When (C) lands on the cushioned floor of the shaft, the rope becomes slack; (W) continues its motion and is finally stopped by a spring bumper.



Doc. 3

Masses: (W) $m_1 = 200$ kg; (C) $m_2 = 100$ kg. Both are considered as particles.
At $t_0 = 0$, (C) is at the height $h = 1.5$ m above the floor of the shaft; the stage is at the height $H = 2.5$ m above this floor.
The stage exerts on (W) a friction force of constant magnitude $f = 100$ N. The rope is inextensible; the rope and the pulley have negligible masses, and the pulley turns without friction. While the rope is taut, (W) and (C) have the same speed v.
The position of (W) is located by its abscissa x on the horizontal axis $(O, \vec{i}\,)$, O being its initial position.
The floor of the shaft is the reference level for gravitational potential energy.

A sensor gives the speed of (W) during the first second (Doc. 4).

t (s)	0	0.2	0.4	0.6	0.8	1.0
v (m/s)	0	0.60	1.20	1.80	2.40	3.00

Doc. 4

Part 1 — The counterweight pulls ($0 \le x \le h$)

1.1 Write the expression of the mechanical energy of the system $\mathcal{S} = [(\text{W}), (\text{C}), \text{rope}, \text{Earth}]$ in terms of m_1, m_2, g, H, h, x and v.

1.2 Is the mechanical energy of $\mathcal{S}$ conserved? Justify.

1.3 By an energetic study, determine the expression of the acceleration a of (W) in terms of m_1, m_2, g and f. Calculate a and specify the nature of the motion of (W).

1.4 Verify the value of a using Doc. 4.

1.5 Determine the instant t_1 at which (C) reaches the floor and the speed v_1 of (W) at this instant. Verify v_1 by an energetic method.

1.6 Calculate the variation of the internal energy of the system $[\mathcal{S}, \text{stage}]$ during this phase.

Part 2 — The wagon alone ($h \le x \le h + D$, with $D = 5.0\ \text{m}$)

(C) stops on the cushion as soon as it touches it; the rope becomes slack.

2.1 Calculate the energy transformed into internal energy when (C) lands.

2.2 By an energetic study of the system [(W), Earth], determine the acceleration a' of (W). Specify the nature of its motion.

2.3 Determine the speed v_2 with which (W) reaches the bumper, and the duration of this phase.

Part 3 — The spring bumper

The bumper behaves as a horizontal spring of negligible mass and of stiffness $k = 1.2 \times 10^4\ \text{N/m}$. At an instant, the spring is compressed by X. The friction of the stage still acts; its maximum value when (W) is at rest is also f.

3.1 Write the expression of the mechanical energy of the system [(W), spring, Earth]. By deriving it with respect to time, determine the expression of the acceleration of (W) in terms of f, k, X and m_1. Specify the nature of the motion.

3.2 Show that the maximum compression of the spring is $X_m = 25\ \text{cm}$.

3.3 Calculate the acceleration of (W) at the instant it touches the bumper and at the maximum compression.

3.4 Show that (W) does not remain at rest at the maximum compression. Determine its speed when it leaves the bumper, then the distance it travels and the time it takes before it stops.

3.5 Show that the total variation of internal energy, from the release to the final stop, is equal to $m_2 g h$. Interpret this result.

— End of the test —